THE EXAM WORKBOOK

# Grade 10 Advanced

## افهم الفكرة. أتقن الحل.

Algebra. Similarity. Trigonometry.

**25** learning outcomes

**100** focused questions

**60** minute online mock

Name / الاسم _____ Class / الصف _____

### A short route to confident practice

READ → REASON → CALCULATE → CHECK. Calculator guides on the website use fx-991EX ClassWiz.

اقرأ القاعدة، حلّ بنفسك، ثم راجع خطواتك. لكل مخرج 4 أسئلة مختارة، الرسومات من الكتاب الأصلي.

### Your topic map


| | | |
|---|---|---|
| Q1 | Polynomial operations | 2 |
| Q2 | FOIL and binomial products | 2 |
| Q3 | Factoring by GCF and grouping | 3 |
| Q4 | Factoring monic trinomials | 3 |
| Q5 | Interior angles of polygons | 4 |
| Q6 | Properties of parallelograms | 5 |
| Q7 | Tests for parallelograms | 6 |
| Q8 | Kites | 7 |
| Q9 | Dilation coordinates | 8 |
| Q10 | Similar polygons | 9 |
| Q11 | Dilation scale factors | 10 |
| Q12 | AA triangle similarity | 11 |
| Q13 | Proportional parts in triangles | 12 |
| Q14 | Parts of similar triangles | 13 |
| Q15 | Geometric mean | 14 |
| Q16 | Pythagorean theorem and converse | 15 |
| Q17 | Special right triangles | 16 |
| Q18 | Right-triangle trigonometry | 17 |
| Q19 | Angles of elevation and depression | 18 |
| Q20 | Area with two sides and an angle | 19 |
| Q21 | Special-product factorization | 20 |
| Q22 | Interior and exterior angle reasoning | 21 |
| Q23 | Solving with similar triangles | 22 |
| Q24 | Distance and midpoint in space | 24 |
| Q25 | Law of Sines and the ambiguous case | 26 |
| KEY | MCQ answers | 28 |
| KEY | Worked written answers | 29 |


Selected from the uploaded Reveal textbook, printed pp. 57–220 and 627–686, aligned to the 25 established learning outcomes. Written tasks and original diagrams come from the book. MCQs include exam-style adaptations and labelled concept/example practice. This is a study selection, not an official exam or a prediction. Use the rounding instructions stated in each original book question. Online mock: 20 MCQs / 60 minutes. Calculator: fx-991EX ClassWiz.

معطيات المقالية من الكتاب، والاختيار من متعدد يتضمن تطبيقات موثقة للمفاهيم والأمثلة. لا يمكن ضمان ورود أسئلة بعينها في الاختبار النهائي.

01 / CHOOSE & CHECK

# Q1 — Q2

## Q1 Polynomial operations

عمليات كثيرات الحدود

**Remember** Order terms by decreasing powers. Combine like terms. When subtracting a polynomial, change the sign of every term in the second polynomial.

### 1.1 MCQ

Which option shows the polynomial in standard form and its leading coefficient?

$$-y^3 + 3y - 3y^2 + 2$$

Reveal · p. 635 · 10 · MCQ adaptation

A $-y^3 - 3y^2 + 3y + 2, \quad 1$

B $-y^3 - 3y^2 + 3y + 2, \quad 2$

C $-y^3 - 3y^2 + 3y + 2, \quad -1$

D $y^3 + 3y^2 - 3y - 2, \quad 1$

### 1.2 MCQ

Which option shows the polynomial in standard form and its leading coefficient?

$$\frac{1}{2}x - 3x^4 + 7$$

Reveal · p. 635 · 13 · MCQ adaptation

A $-3x^4 + \frac{x}{2} + 7, \quad 2$

B $-3x^4 + \frac{x}{2} + 7, \quad -3$

C $3x^4 - \frac{x}{2} - 7, \quad 3$

D $-3x^4 + \frac{x}{2} + 7, \quad 3$

### 1.3 MCQ

Which option gives the sum or difference?

$$(2h^2 - 5h) + (7h - 3h^2)$$

Reveal · p. 635 · 22 · MCQ adaptation

A $-h^2 + 2h$

B $2h$

C $h^2 + 2h$

D $-h^2$

### 1.4 MCQ

Which option gives the sum or difference?

$$(2x + 3x^2) - (7 - 8x^2)$$

Reveal · p. 635 · 26 · MCQ adaptation

A $11x^2 + 2x$

B $11x^2 + 2x + 7$

C $11x^2 - 7$

D $(11x^2 + 2x) - 7$

## Q2 FOIL and binomial products

ضرب ثنائيي الحد بطريقة FOIL

**Remember** (a + b)(c + d) = ac + ad + bc + bd. Combine the two middle products when they are like terms.

### 2.1 MCQ

Which option gives the product?

$$(c + 3)(3c - 5)$$

Reveal · p. 649 · 1 · MCQ adaptation

A $3c^2 + 4c$

B $(3c^2 + 4c) - 15$

C $3c^2 + 4c + 15$

D $4c - 15$

### 2.2 MCQ

Which option gives the product?

$$(g + 10)(2g - 5)$$

Reveal · p. 649 · 2 · MCQ adaptation

A $(2g^2 + 15g) - 50$

B $2g^2 + 15g$

C $2g^2 + 15g + 50$

D $15g - 50$

### 2.3 MCQ

Which option gives the product?

$$(5a + 3)(6a + 5)$$

Reveal · p. 649 · 3 · MCQ adaptation

A $30a^2 + 43a$

B $30a^2 + 43a - 15$

C $43a + 15$

D $(30a^2 + 43a) + 15$

### 2.4 MCQ

Which option gives the product?

$$(4x + 1)(6x + 3)$$

Reveal · p. 649 · 4 · MCQ adaptation

A $24x^2 + 18x$

B $24x^2 + 18x - 3$

C $(24x^2 + 18x) + 3$

D $24x^2 + 3$

01 / CHOOSE & CHECK

# Q3 — Q4

## Q3 Factoring by GCF and grouping

التحليل بالعامل المشترك والتجميع

**Remember** Factor out the greatest common factor first. For grouping, arrange terms so both groups contain the same binomial factor.

**3.1** MCQ

Which option is the complete factorisation of the expression?

$$fg + 4f - 5g - 20$$

Reveal · p. 665 · 11 · MCQ adaptation

A $(f-5)(g+4)$
B $(-f-5)(g+4)$
C $(f+5)(g+4)$
D $(4-g)(f-5)$

**3.2** MCQ

Which option is the complete factorisation of the expression?

$$hj - 2h + 5j - 10$$

Reveal · p. 665 · 13 · MCQ adaptation

A $(5-h)(j-2)$
B $(h-5)(j-2)$
C $(h+5)(-j-2)$
D $(h+5)(j-2)$

**3.3** MCQ

Which option is the complete factorisation of the expression?

$$xy - 2x + y - 2$$

Reveal · p. 665 · 14 · MCQ adaptation

A $(1-x)(y-2)$
B $(x-1)(y-2)$
C $(x+1)(y-2)$
D $(x+1)(-y-2)$

**3.4** MCQ

Which option is the complete factorisation of the expression?

$$pv + 8p + 12v + 96$$

Reveal · p. 665 · 20 · MCQ adaptation

A $(12-p)(v+8)$
B $(p+12)(v+8)$
C $(p-12)(v+8)$
D $(8-v)(p+12)$

## Q4 Factoring monic trinomials

تحليل ثلاثيات الحدود ذات المعامل الرئيسي 1

**Remember** Find integers m and n with m + n = b and mn = c. Then x² + bx + c = (x + m)(x + n).

**4.1** MCQ

Which option is the complete factorisation of the expression?

$$x^2 + 17x + 42$$

Reveal · p. 671 · 1 · MCQ adaptation

A $(3-x)(x+14)$
B $(x-3)(x+14)$
C $(14-x)(x+3)$
D $(x+3)(x+14)$

**4.2** MCQ

Which option is the complete factorisation of the expression?

$$y^2 - 17y + 72$$

Reveal · p. 671 · 2 · MCQ adaptation

A $(-y-9)(y-8)$
B $(y-8)(y+9)$
C $(y-9)(y-8)$
D $(-y-8)(y-9)$

**4.3** MCQ

Which option is the complete factorisation of the expression?

$$a^2 + 8a - 48$$

Reveal · p. 671 · 3 · MCQ adaptation

A $(-a-4)(a+12)$
B $(a-4)(a+12)$
C $(a+4)(a+12)$
D $(12-a)(a-4)$

**4.4** MCQ

Which option is the complete factorisation of the expression?

$$n^2 - 2n - 35$$

Reveal · p. 671 · 4 · MCQ adaptation

A $(n-7)(n+5)$
B $(-n-7)(n+5)$
C $(n+5)(n+7)$
D $(5-n)(n-7)$

01 / CHOOSE & CHECK

# Q5

## Q5 Interior angles of polygons
الزوايا الداخلية للمضلعات

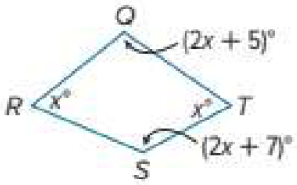
**Remember** For a simple n-sided polygon, the interior angle sum is (n − 2)180°. For a regular polygon, each interior angle is (n − 2)180°/n.

### 5.1 MCQ

Which option gives the interior angles in the stated vertex order?

Reveal · p. 63 · 1 · MCQ adaptation

1.




A $(Q, R, S, T) = (117, 56, 119, 56)^\circ$

B $(Q, R, S, T) = (125, 60, 127, 60)^\circ$

C $(Q, R, S, T) = (121, 58, 123, 58)^\circ$

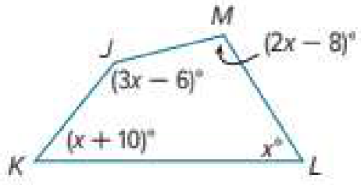
D $(Q, R, S, T) = (120, 60, 120, 60)^\circ$

### 5.2 MCQ

Which option gives the interior angles in the stated vertex order?

Reveal · p. 63 · 2 · MCQ adaptation

2.




A $(J, K, L, M) = (150, 52, 62, 96)^\circ$

B $(J, K, L, M) = (150, 62, 52, 96)^\circ$

C $(J, K, L, M) = (144, 60, 50, 92)^\circ$

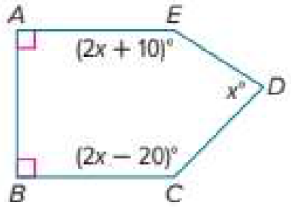
D $(J, K, L, M) = (156, 64, 54, 100)^\circ$

### 5.3 MCQ

Which option gives the interior angles in the stated vertex order?

Reveal · p. 63 · 3 · MCQ adaptation

3.




A $(A, B, C, D, E) = (90, 90, 128, 74, 158)^\circ$

B $(A, B, C, D, E) = (90, 90, 140, 80, 170)^\circ$

C $(A, B, C, D, E) = (90, 90, 124, 72, 154)^\circ$

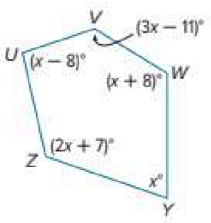
D $(A, B, C, D, E) = (90, 90, 158, 74, 128)^\circ$

### 5.4 MCQ

Which option gives the interior angles in the stated vertex order?

Reveal · p. 63 · 4 · MCQ adaptation

4.




A $(U, V, W, Y, Z) = (62, 199, 78, 70, 147)^\circ$

B $(U, V, W, Y, Z) = (68, 217, 84, 76, 159)^\circ$

C $(U, V, W, Y, Z) = (60, 143, 76, 68, 193)^\circ$

D $(U, V, W, Y, Z) = (60, 193, 76, 68, 143)^\circ$

01 / CHOOSE & CHECK

# Q6

## Q6 Properties of parallelograms

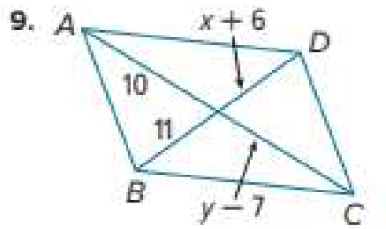
خصائص متوازيات الأضلاع

**Remember** Opposite sides and opposite angles are congruent. Consecutive angles are supplementary. A diagonal divides a parallelogram into two congruent triangles.

### 6.1 MCQ

Which values of the variables satisfy the parallelogram shown?

Reveal · p. 72 · 9 · MCQ adaptation




A $(x, y) = (16, 3)$

B $(x, y) = (5, 17)$

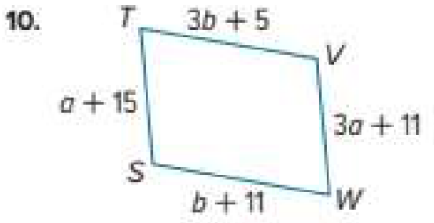
C $(x, y) = (11, 10)$

D $(x, y) = (17, 5)$

### 6.2 MCQ

Which values of the variables satisfy the parallelogram shown?

Reveal · p. 72 · 10 · MCQ adaptation




A $(a, b) = (2, 3)$

B $(a, b) = (3, 2)$

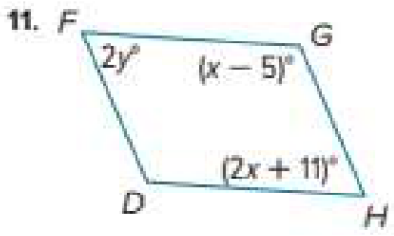
C $(a, b) = (13, 6)$

D $(a, b) = (4, 3)$

### 6.3 MCQ

Which values of the variables satisfy the parallelogram shown?

Reveal · p. 72 · 11 · MCQ adaptation




A $(x, y) = (58, 29)$

B $(x, y) = (60, 60)$

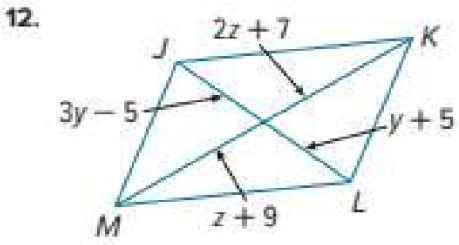
C $(x, y) = (53, 58)$

D $(x, y) = (58, \frac{127}{2})$

### 6.4 MCQ

Which values of the variables satisfy the parallelogram shown?

Reveal · p. 72 · 12 · MCQ adaptation




A $(y, z) = (3, 2)$

B $(y, z) = (5, 9)$

C $(y, z) = (5, 2)$

D $(y, z) = (2, 5)$

01 / CHOOSE & CHECK

# Q7

## Q7 Tests for parallelograms
اختبارات متوازي الأضلاع

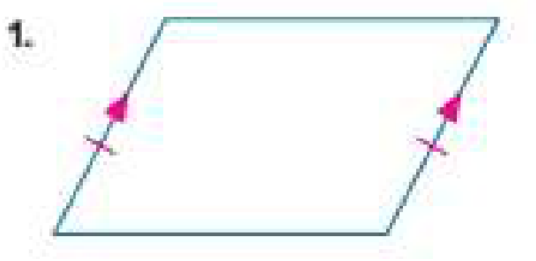
**Remember** Sufficient tests include both pairs of opposite sides congruent, both pairs of opposite angles congruent, diagonals bisecting each other, or one pair of opposite sides both parallel and congruent.

### 7.1 MCQ

Determine whether the markings guarantee a parallelogram. Choose the correct justification.

Reveal · p. 79 · 1 · MCQ adaptation

1.



A Yes: one pair of opposite sides is both parallel and congruent.

B Yes: the diagonals are marked perpendicular.

C No: a parallelogram cannot have these markings.

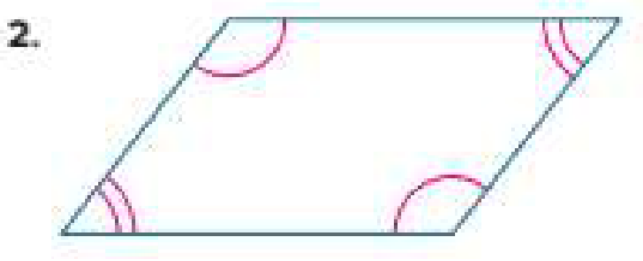
D Yes: any one pair of congruent sides is sufficient.

### 7.2 MCQ

Determine whether the markings guarantee a parallelogram. Choose the correct justification.

Reveal · p. 79 · 2 · MCQ adaptation

2.



A Yes: the diagonals are marked perpendicular.

B No: a parallelogram cannot have these markings.

C Yes: any one pair of congruent sides is sufficient.

D Yes: both pairs of opposite angles are congruent.

### 7.5 MCQ

Which condition guarantees a quadrilateral is a parallelogram?

Reveal · p. 79 · Application of 1 · Concept application

A Both pairs of opposite sides are congruent.

B Exactly one right angle.

C Only one pair of opposite sides is congruent.

D Its diagonals have equal length.

### 7.6 MCQ

Which extra condition makes a quadrilateral with one pair of parallel opposite sides a parallelogram?

Reveal · p. 79 · Application of 1 · Concept application

A One interior angle is acute.

B The diagonals are congruent.

C That same pair is congruent.

D The other pair is congruent.

01 / CHOOSE & CHECK

# Q8

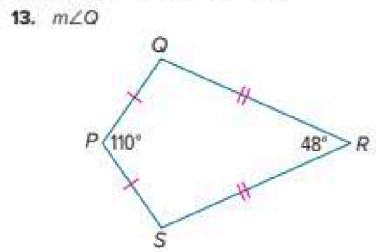
## Q8 Kites
الطائرات الورقية

**Remember** In a kite with AB = AD and CB = CD, AC is the symmetry diagonal. AC is perpendicular to BD and bisects BD. The opposite angles at B and D are congruent.

### 8.1 MCQ

Which option gives the requested angle measure in the kite?

Reveal · p. 106 · 13 · MCQ adaptation

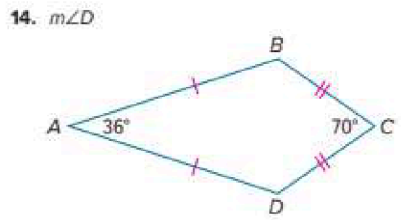



A 202
B 79
C 158
D 101

### 8.2 MCQ

Which option gives the requested angle measure in the kite?

Reveal · p. 106 · 14 · MCQ adaptation

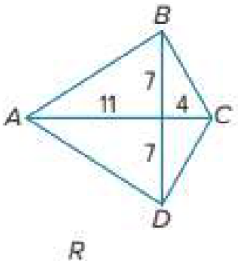



A 254
B 53
C 127
D 106

### 8.3 MCQ

Which option gives the requested side in simplest radical form, then the perimeter to the nearest tenth?

Reveal · p. 106 · 15 · MCQ adaptation

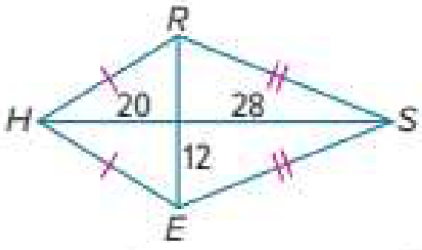



A $BC = \sqrt{65}, \quad P \approx 58.5$
B $BC = \sqrt{65}, \quad P \approx 42.2$
C $BC = \sqrt{170}, \quad P \approx 42.2$
D $BC = \sqrt{65}, \quad P \approx 21.1$

### 8.4 MCQ

Which option gives the requested side in simplest radical form, then the perimeter to the nearest tenth?

Reveal · p. 106 · 16 · MCQ adaptation




A $RH = 4\sqrt{34}, \quad P \approx 107.6$
B $RH = 4\sqrt{58}, \quad P \approx 107.6$
C $RH = 4\sqrt{34}, \quad P \approx 53.8$
D $RH = 4\sqrt{34}, \quad P \approx 96.0$

01 / CHOOSE & CHECK

# Q9

Q9

## Dilation coordinates

إحداثيات التمدد

**Remember** For a dilation centered at the origin, (x, y) maps to (kx, ky). Use image/original to find k, retaining the signs of coordinates.

### 9.1 MCQ

Graph the image after a dilation centred at the origin. Choose the complete set of image vertices.

$$F(-10,4) \quad G(-4,4) \quad H(-4,-8)$$

$$k = \frac{1}{4}$$

Reveal · p. 120 · 13 · MCQ adaptation

A $F' = (\frac{5}{2}, -1)$, $G' = (1, -1)$, $H' = (1, 2)$

B $F' = (-\frac{25}{2}, 5)$, $G' = (-5, 5)$, $H' = (-5, -10)$

C $F' = (-\frac{5}{2}, 1)$, $G' = (-1, 1)$, $H' = (-1, -2)$

D $F' = (-40, 16)$, $G' = (-16, 16)$, $H' = (-16, -32)$

### 9.2 MCQ

Graph the image after a dilation centred at the origin. Choose the complete set of image vertices.

$$X(2,-1) \quad Y(-6,4) \quad Z(-2,-5)$$

$$k = \frac{5}{4}$$

Reveal · p. 120 · 14 · MCQ adaptation

A $X' = (\frac{9}{2}, -\frac{9}{4})$, $Y' = (-\frac{27}{2}, 9)$, $Z' = (-\frac{9}{2}, -\frac{45}{4})$

B $X' = (\frac{5}{2}, -\frac{5}{4})$, $Y' = (-\frac{15}{2}, 5)$, $Z' = (-\frac{5}{2}, -\frac{25}{4})$

C $X' = (\frac{8}{5}, -\frac{4}{5})$, $Y' = (-\frac{24}{5}, \frac{16}{5})$, $Z' = (-\frac{8}{5}, -4)$

D $X' = (-\frac{5}{2}, \frac{5}{4})$, $Y' = (\frac{15}{2}, -5)$, $Z' = (\frac{5}{2}, \frac{25}{4})$

### 9.3 MCQ

Graph the image after a dilation centred at the origin. Choose the complete set of image vertices.

$$M(4,6) \quad N(-6,2) \quad P(0,-8)$$

$$k = \frac{3}{4}$$

Reveal · p. 120 · 15 · MCQ adaptation

A $M' = (3, \frac{9}{2})$, $N' = (-\frac{9}{2}, \frac{3}{2})$, $P' = (0, -6)$

B $M' = (\frac{16}{3}, 8)$, $N' = (-8, \frac{8}{3})$, $P' = (0, -\frac{32}{3})$

C $M' = (-3, -\frac{9}{2})$, $N' = (\frac{9}{2}, -\frac{3}{2})$, $P' = (0, 6)$

D $M' = (7, \frac{21}{2})$, $N' = (-\frac{21}{2}, \frac{7}{2})$, $P' = (0, -14)$

### 9.4 MCQ

Graph the image after a dilation centred at the origin. Choose the complete set of image vertices.

$$R(-2,6) \quad S(0,-1) \quad T(-5,3)$$

$$k = \frac{3}{2}$$

Reveal · p. 120 · 16 · MCQ adaptation

A $R' = (-\frac{4}{3}, 4)$, $S' = (0, -\frac{2}{3})$, $T' = (-\frac{10}{3}, 2)$

B $R' = (3, -9)$, $S' = (0, \frac{3}{2})$, $T' = (\frac{15}{2}, -\frac{9}{2})$

C $R' = (-5, 15)$, $S' = (0, -\frac{5}{2})$, $T' = (-\frac{25}{2}, \frac{15}{2})$

D $R' = (-3, 9)$, $S' = (0, -\frac{3}{2})$, $T' = (-\frac{15}{2}, \frac{9}{2})$

01 / CHOOSE & CHECK

# Q10

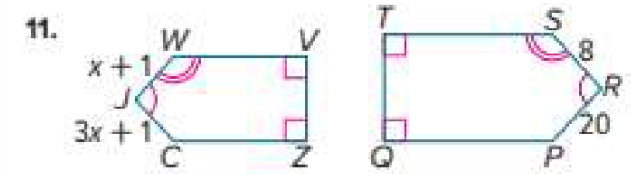
## Q10 Similar polygons
المضلعات المتشابهة

**Remember** Corresponding angles of similar polygons are congruent. Corresponding side lengths have one common ratio. Match vertices in the stated order.

**10.1** MCQ

Each pair of polygons is similar. Find x.

Reveal · p. 128 · 11 · MCQ adaptation

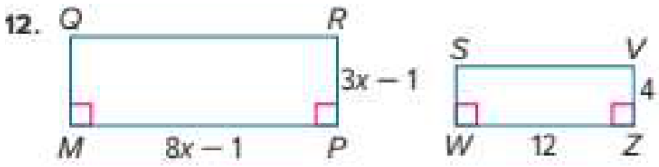



A 6
B 3
C 4
D 2

**10.2** MCQ

Each pair of polygons is similar. Find x.

Reveal · p. 128 · 12 · MCQ adaptation

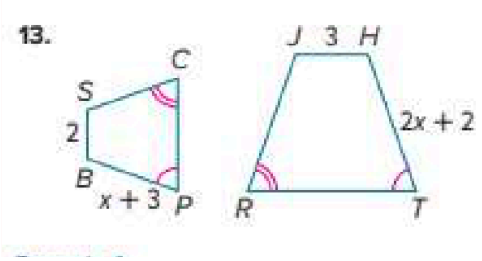



A 2
B 3
C 1
D 5

**10.3** MCQ

Each pair of polygons is similar. Find x.

Reveal · p. 128 · 13 · MCQ adaptation

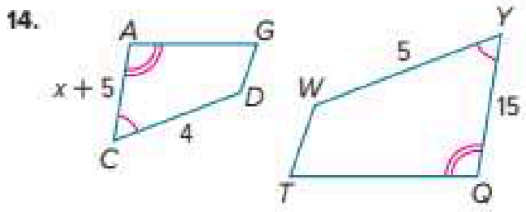



A 6
B 4
C 8
D 5

**10.4** MCQ

Each pair of polygons is similar. Find x.

Reveal · p. 128 · 14 · MCQ adaptation




A 6
B 10
C 7
D 8

01 / CHOOSE & CHECK

# Q11

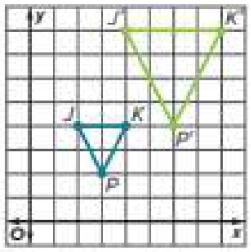
## Q11 Dilation scale factors
معاملات التمدد

**Remember** Scale factor = image length/original length. For positive k, 0 < k < 1 gives a reduction and k > 1 gives an enlargement. A dilation preserves angle measures.

**11.1** MCQ

Which option gives the scale factor from the original polygon to its primed image?

Reveal · p. 120 · 10 · MCQ adaptation

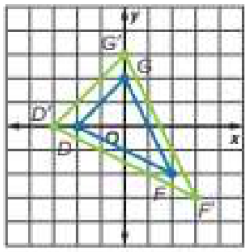



A $2$
B $\frac{1}{2}$
C $-2$
D $3$

**11.2** MCQ

Which option gives the scale factor from the original polygon to its primed image?

Reveal · p. 120 · 11 · MCQ adaptation

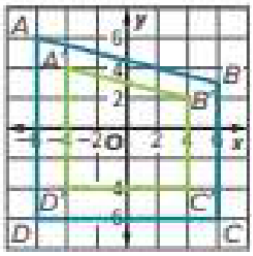



A $\frac{2}{3}$
B $-\frac{3}{2}$
C $\frac{5}{2}$
D $\frac{3}{2}$

**11.3** MCQ

Which option gives the scale factor from the original polygon to its primed image?

Reveal · p. 120 · 17 · MCQ adaptation

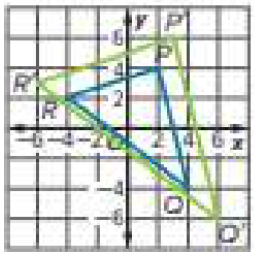



A $-\frac{2}{3}$
B $\frac{5}{3}$
C $\frac{2}{3}$
D $\frac{3}{2}$

**11.4** MCQ

Which option gives the scale factor from the original polygon to its primed image?

Reveal · p. 120 · 18 · MCQ adaptation




A $\frac{5}{2}$
B $\frac{3}{2}$
C $\frac{2}{3}$
D $-\frac{3}{2}$

01 / CHOOSE & CHECK

# Q12

## Q12 AA triangle similarity

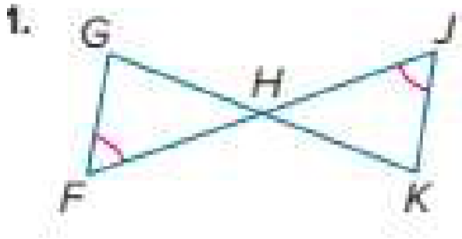
تشابه المثلثات بزاويتين

**Remember** Two pairs of congruent corresponding angles establish triangle similarity. Parallel lines and vertical angles can supply those angle pairs.

### 12.1 MCQ

Determine whether the triangles are similar. Choose a valid explanation.

Reveal · p. 133 · 1 · MCQ adaptation

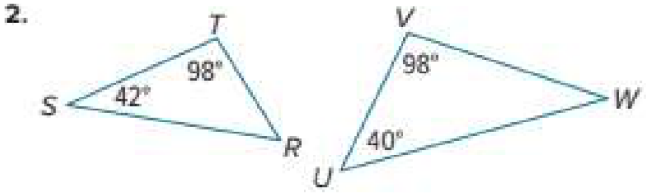



- A Not similar: equal vertical angles prevent similarity.
- B Not enough information: no angle pair is known.
- C Similar by SSS: three side lengths are given.
- D Similar by AA: the marked angles and vertical angles are equal.

### 12.2 MCQ

Determine whether the triangles are similar. Choose a valid explanation.

Reveal · p. 133 · 2 · MCQ adaptation




- A Not enough information: the third angles cannot be found.
- B Similar by SSS only.
- C Similar by AA: their angle sets are 42°, 98°, 40°.
- D Not similar: one triangle has 42° and the other has 40°.

### 12.5 MCQ

Two angles of one triangle equal two angles of a second triangle. Which conclusion follows?

Reveal · p. 133 · Application of 1 · Concept application

- A The triangles must be congruent.
- B All corresponding sides are equal.
- C The triangles cannot be similar.
- D The triangles are similar by AA.

### 12.6 MCQ

Which information alone is insufficient to prove two triangles similar?

Reveal · p. 133 · Application of 1 · Concept application

- A Only one pair of equal angles.
- B Two pairs of equal angles.
- C All three corresponding side ratios equal.
- D Two corresponding side ratios equal and the included angles equal.

01 / CHOOSE & CHECK

# Q13

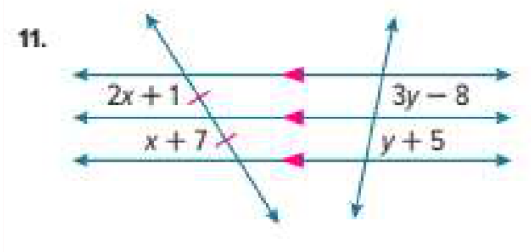
## Q13 Proportional parts in triangles
الأجزاء المتناسبة في المثلثات

**Remember** A line parallel to one side of a triangle divides the other two sides proportionally. A triangle midsegment is parallel to the third side and half its length.

### 13.1 MCQ

Which option gives the labelled variable(s), using the parallel-line and midpoint markings?

Reveal · p. 146 · 11 · MCQ adaptation

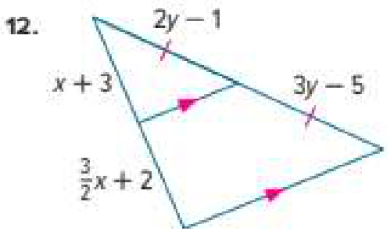



A $(x, y) = (7, 6)$
B $(x, y) = (13, 6)$
C $(x, y) = (6, \frac{13}{2})$
D $(x, y) = (6, 13)$

### 13.2 MCQ

Which option gives the labelled variable(s), using the parallel-line and midpoint markings?

Reveal · p. 146 · 12 · MCQ adaptation

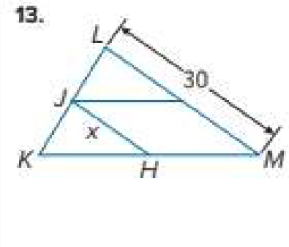



A $(x, y) = (3, 4)$
B $(x, y) = (2, 4)$
C $(x, y) = (4, 2)$
D $(x, y) = (2, 3)$

### 13.3 MCQ

Which option gives the labelled variable(s), using the parallel-line and midpoint markings?

Reveal · p. 146 · 13 · MCQ adaptation

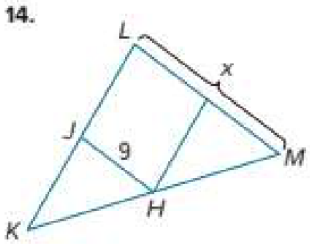



A $x = 15$
B $x = 30$
C $x = 60$
D $x = 20$

### 13.4 MCQ

Which option gives the labelled variable(s), using the parallel-line and midpoint markings?

Reveal · p. 146 · 14 · MCQ adaptation




A $x = 9$
B $x = 4.5$
C $x = 27$
D $x = 18$

01 / CHOOSE & CHECK

# Q14

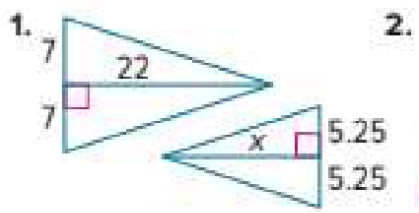
## Q14 Parts of similar triangles
أجزاء المثلثات المتشابهة

**Remember** Corresponding altitudes, medians, and angle-bisector lengths of similar triangles have the same ratio as corresponding sides. Within one triangle, an angle bisector divides the opposite side in the ratio of the adjacent sides.

**14.1** MCQ

The triangles are similar. Find x.

Reveal · p. 153 · 1 · MCQ adaptation

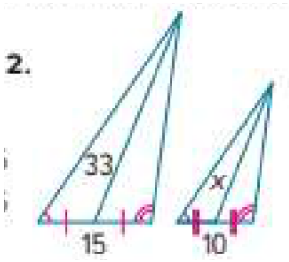



A $\frac{88}{3}$ B $\frac{33}{2}$
C $\frac{33}{4}$ D 33

**14.2** MCQ

The triangles are similar. Find x.

Reveal · p. 153 · 2 · MCQ adaptation

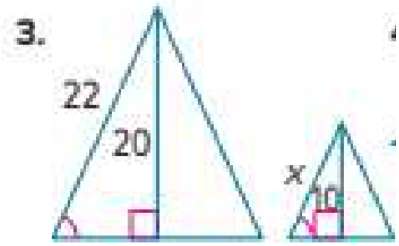



A 22 B $\frac{99}{2}$
C 33 D 11

**14.3** MCQ

The triangles are similar. Find x.

Reveal · p. 153 · 3 · MCQ adaptation

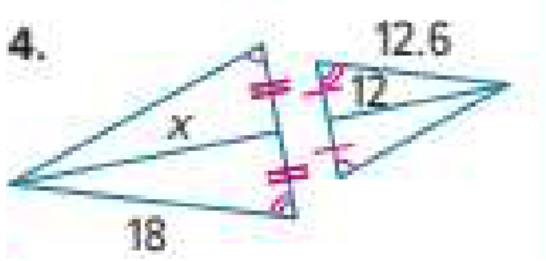



A 44 B 22
C $\frac{11}{2}$ D 11

**14.4** MCQ

The triangles are similar. Find x.

Reveal · p. 153 · 4 · MCQ adaptation




A $\frac{42}{5}$ B $\frac{60}{7}$
C $\frac{120}{7}$ D $\frac{240}{7}$

01 / CHOOSE & CHECK

# Q15

## Q15 Geometric mean
الوسط الهندسي

**Remember** The geometric mean of positive a and b is √(ab). For a right triangle with altitude h to a hypotenuse split into p and q, $h^2 = pq$. A leg squared equals the whole hypotenuse times its adjacent hypotenuse segment.

### 15.1 MCQ

What is the positive geometric mean of the two numbers?

$4 \quad \text{and} \quad 6$

Reveal · p. 165 · 1 · MCQ adaptation

A $2\sqrt{6}$
B $5$
C $24$
D $-2\sqrt{6}$

### 15.2 MCQ

What is the positive geometric mean of the two numbers?

$\frac{1}{2} \quad \text{and} \quad 2$

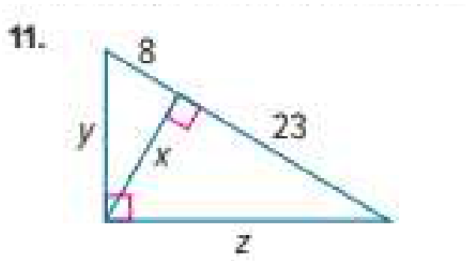
Reveal · p. 165 · 2 · MCQ adaptation

A $\frac{5}{4}$
B $\frac{5}{2}$
C $-1$
D $1$

### 15.3 MCQ

Which values of x, y and z are correct, in simplest exact form?

Reveal · p. 165 · 11 · MCQ adaptation

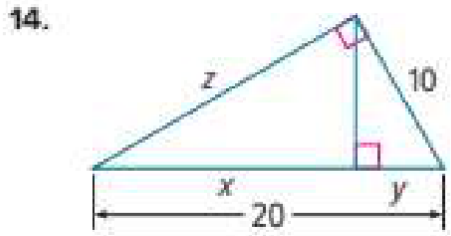



A $(x, y, z) = (\sqrt{184}, \sqrt{713}, \sqrt{248})$
B $(x, y, z) = (\sqrt{248}, \sqrt{184}, \sqrt{713})$
C $(x, y, z) = (2\sqrt{46}, 2\sqrt{62}, \sqrt{713})$
D $(x, y, z) = (\sqrt{31}, \sqrt{184}, \sqrt{248})$

### 15.4 MCQ

Which values of x, y and z are correct, in simplest exact form?

Reveal · p. 165 · 14 · MCQ adaptation




A $(x, y, z) = (10, 10, 10\sqrt{2})$
B $(x, y, z) = (15, 5, 10\sqrt{3})$
C $(x, y, z) = (5, 15, 10\sqrt{3})$
D $(x, y, z) = (15, 5, 5\sqrt{3})$

01 / CHOOSE & CHECK

# Q16

## Q16 Pythagorean theorem and converse

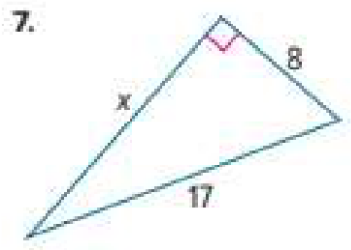
نظرية فيثاغورس وعكسها

**Remember** For a right triangle, $a^2 + b^2 = c^2$ with c the hypotenuse. For valid side lengths and longest side c, compare $c^2$ with $a^2 + b^2$ to classify the triangle.

**16.1** MCQ

Use a Pythagorean triple to find x.

Reveal · p. 171 · 7 · MCQ adaptation

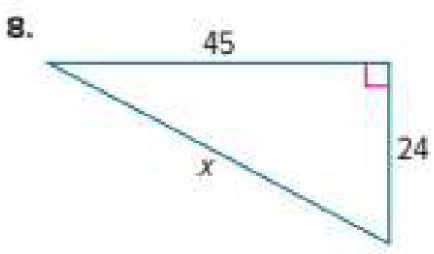



A $\sqrt{353}$
B 9
C 225
D 15

**16.2** MCQ

Use a Pythagorean triple to find x.

Reveal · p. 171 · 8 · MCQ adaptation




A 69
B 21
C 51
D $\sqrt{1449}$

**16.5** MCQ

How is the triangle with these side lengths classified?

$\sqrt{5}, \sqrt{12}, \sqrt{13}$

Reveal · p. 171 · Application of 7 · Concept application

A Obtuse
B Not a triangle
C Acute
D Right

**16.6** MCQ

How is the triangle with these side lengths classified?

$2, \sqrt{8}, \sqrt{12}$

Reveal · p. 171 · Application of 7 · Concept application

A Obtuse
B Not a triangle
C Right
D Acute

01 / CHOOSE & CHECK

# Q17

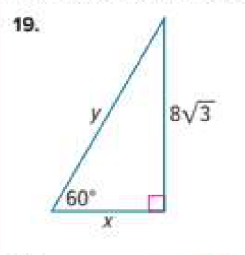
## Q17 Special right triangles
المثلثات القائمة الخاصة

**Remember** 45°–45°–90° side ratio is 1:1:√2. In a 30°–60°–90° triangle, sides opposite 30°, 60°, and 90° have ratio 1:√3:2.

**17.1** MCQ

Which values of x and y are correct, in simplest exact form?

Reveal · p. 184 · 19 · MCQ adaptation

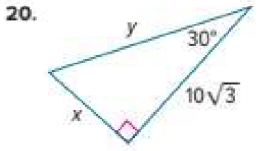



A $(x, y) = (8\sqrt{3}, 16)$
B $(x, y) = (8, 8\sqrt{3})$
C $(x, y) = (8, 16)$
D $(x, y) = (16, 8)$

**17.2** MCQ

Which values of x and y are correct, in simplest exact form?

Reveal · p. 184 · 20 · MCQ adaptation




A $(x, y) = (10, 10\sqrt{3})$
B $(x, y) = (10, 20)$
C $(x, y) = (20, 10)$
D $(x, y) = (10\sqrt{3}, 20)$

**17.5** MCQ

A 45°–45°–90° triangle has legs of length 6. What is its hypotenuse?

Reveal · p. 184 · Application of 19 · Concept application

A $3\sqrt{2}$
B $6\sqrt{2}$
C $12$
D $6$

**17.6** MCQ

A 45°–45°–90° triangle has hypotenuse 10. What is each leg?

Reveal · p. 184 · Application of 19 · Concept application

A $5\sqrt{2}$
B $10\sqrt{2}$
C $5$
D $20$

01 / CHOOSE & CHECK

# Q18

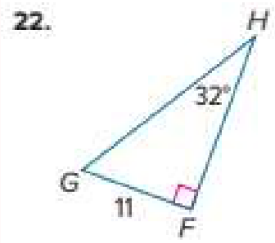
## Q18 Right-triangle trigonometry
النسب المثلثية في المثلث القائم

**Remember** sin θ = opposite/hypotenuse, cos θ = adjacent/hypotenuse, tan θ = opposite/adjacent. Use inverse ratios for angles. Set a scientific calculator to degrees.

### 18.1 MCQ

Solve the right triangle. Round sides to the nearest tenth and acute angles to the nearest degree.

Reveal · p. 192 · 22 · MCQ adaptation




A $\angle G = 58^\circ$, $FH \approx 6.9$, $GH \approx 13.0$

B $\angle G = 58^\circ$, $FH \approx 17.6$, $GH \approx 20.8$

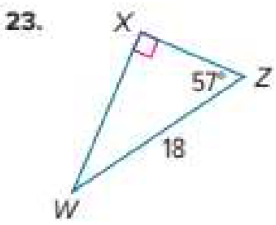
C $\angle G = 58^\circ$, $FH \approx 20.8$, $GH \approx 17.6$

D $\angle G = 32^\circ$, $FH \approx 17.6$, $GH \approx 20.8$

### 18.2 MCQ

Solve the right triangle. Round sides to the nearest tenth and acute angles to the nearest degree.

Reveal · p. 192 · 23 · MCQ adaptation




A $\angle W = 33^\circ$, $WX \approx 15.1$, $XZ \approx 9.8$

B $\angle W = 33^\circ$, $WX \approx 9.8$, $XZ \approx 15.1$

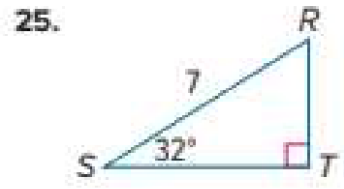
C $\angle W = 57^\circ$, $WX \approx 15.1$, $XZ \approx 9.8$

D $\angle W = 33^\circ$, $WX \approx 21.5$, $XZ \approx 33.1$

### 18.3 MCQ

Solve the right triangle. Round sides to the nearest tenth and acute angles to the nearest degree.

Reveal · p. 192 · 25 · MCQ adaptation




A $\angle R = 58^\circ$, $ST \approx 3.7$, $RT \approx 5.9$

B $\angle R = 32^\circ$, $ST \approx 5.9$, $RT \approx 3.7$

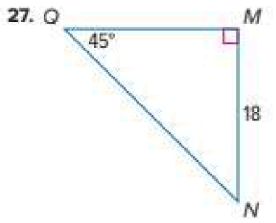
C $\angle R = 58^\circ$, $ST \approx 8.3$, $RT \approx 13.2$

D $\angle R = 58^\circ$, $ST \approx 5.9$, $RT \approx 3.7$

### 18.4 MCQ

Solve the right triangle. Round sides to the nearest tenth and acute angles to the nearest degree.

Reveal · p. 192 · 27 · MCQ adaptation




A $\angle N = 45^\circ$, $QM \approx 25.5$, $QN = 18$

B $\angle N = 90^\circ$, $QM = 18$, $QN \approx 25.5$

C $\angle N = 45^\circ$, $QM = 18$, $QN \approx 25.5$

D $\angle N = 45^\circ$, $QM = 18$, $QN = 36$

01 / CHOOSE & CHECK

# Q19

## Q19 Angles of elevation and depression

زوايا الارتفاع والانخفاض

**Remember** Angles of elevation and depression are measured from horizontal lines. Use vertical height differences, including eye height when given.

**19.1** MCQ

A lighthouse light is 30 m above sea level. Sailors see it at a 25° angle of elevation. How far is the ship from shore, to the nearest metre?

Reveal · p. 199 · 1 · MCQ adaptation

A 64
B 14
C 71
D 33

**19.2** MCQ

A water tower is 110 ft from a student and is 32.5 ft high. The student's eye level is 6 ft. Find the angle of elevation to the nearest tenth of a degree.

Reveal · p. 199 · 2 · MCQ adaptation

A 16.5
B 76.5
C 14.1
D 13.5

**19.3** MCQ

A ladder reaches a 30-ft roof and makes a 55° angle with the ground. How far is its foot from the wall, to the nearest foot?

Reveal · p. 199 · 3 · MCQ adaptation

A 43
B 37
C 21
D 25

**19.4** MCQ

Nabuko is 60 m from a cliff with bridges 100 m high. Her line of sight is 5 m above the ground. Find the elevation angle to the nearest tenth of a degree.

Reveal · p. 199 · 4 · MCQ adaptation

A 59.0
B 57.7
C 32.3
D 46.4

01 / CHOOSE & CHECK

# Q20

## Q20 Area with two sides and an angle

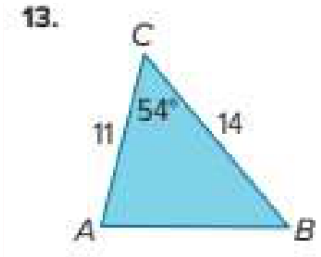
المساحة بضلعين والزاوية المحصورة

**Remember** Area = ½ab sin C, where C is the included angle between sides a and b. If two other angles are given, find the included angle using the 180° sum.

**20.1** MCQ

Use trigonometry to find the triangle's area to the nearest tenth.

Reveal · p. 200 · 13 · MCQ adaptation

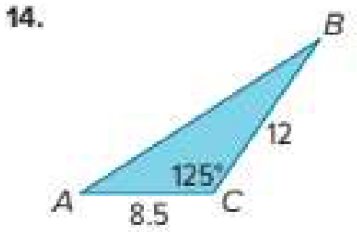



A 124.6
B 45.3
C 77.0
D 62.3

**20.2** MCQ

Use trigonometry to find the triangle's area to the nearest tenth.

Reveal · p. 200 · 14 · MCQ adaptation

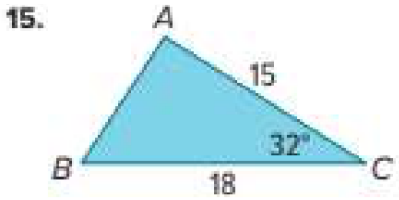



A 83.6
B 51.0
C 41.8
D 29.3

**20.3** MCQ

Use trigonometry to find the triangle's area to the nearest tenth.

Reveal · p. 200 · 15 · MCQ adaptation

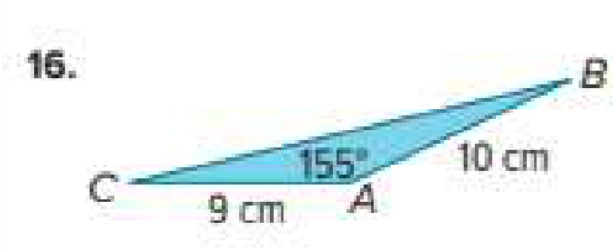



A 143.1
B 71.5
C 114.5
D 135.0

**20.4** MCQ

Use trigonometry to find the triangle's area to the nearest tenth.

Reveal · p. 200 · 16 · MCQ adaptation




A 19.0
B 38.0
C 40.8
D 45.0

02 / WRITE THE STEPS

# Q21 · Written practice 1

Q21

## Special-product factorization

التحليل باستعمال المتطابقات الخاصة

**Remember** $A^2 - B^2 = (A - B)(A + B)$. $A^2 \pm 2AB + B^2 = (A \pm B)^2$. Extract the GCF before applying a special-product pattern.

**21.1** FRQ · show your working

Factor completely using integers.

$$-72p^2 + 8x^2$$

Reveal · p. 680 · 35

**(a)** Show your factorisation and check by expansion.

**21.2** FRQ · show your working

Factor completely using integers.

$$16b^2 - 100$$

Reveal · p. 680 · 38

**(a)** Show your factorisation and check by expansion.

**21.3** FRQ · show your working

Determine whether this is a perfect-square trinomial; if so, factor it.

$$9z^2 - 6z + 1$$

Reveal · p. 681 · 51

**(a)** Show your factorisation and check by expansion.

**21.4** FRQ · show your working

Determine whether this is a perfect-square trinomial; if so, factor it.

$$36x^2 - 60x + 25$$

Reveal · p. 681 · 52

**(a)** Show your factorisation and check by expansion.

02 / WRITE THE STEPS

# Q22 · Written practice 1

## Q22 Interior and exterior angle reasoning

الاستدلال بالزوايا الداخلية والخارجية

**Remember** For a convex polygon, diagonals from one vertex form n − 2 triangles. Interior sum = (n − 2)180°. One exterior angle at each vertex, taken consistently around the polygon, sums to 360°.

**22.1** FRQ · show your working

The given angle is an interior angle of a regular polygon. Find the number of sides.

$144°$

Reveal · p. 64 · 9

**(a)** Show an equation and solve for n.

**22.2** FRQ · show your working

The given angle is an interior angle of a regular polygon. Find the number of sides.

$160°$

Reveal · p. 64 · 11

**(a)** Show an equation and solve for n.

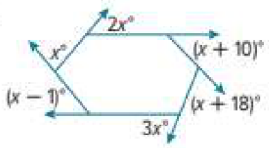
**22.3** FRQ · show your working

Find x in the diagram.

Reveal · p. 64 · 15

**(a)** Show how you use the exterior-angle sum.

15.



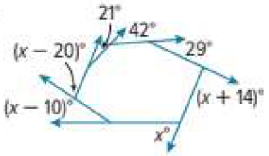

**22.4** FRQ · show your working

Find x in the diagram.

Reveal · p. 64 · 18

**(a)** Show how you use the exterior-angle sum.

18.



02 / WRITE THE STEPS

# Q23 · Written practice 1

## Q23 Solving with similar triangles

حل المسائل باستعمال المثلثات المتشابهة

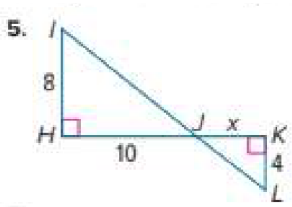
**Remember** Establish similarity by AA, SAS, or SSS. Write corresponding vertices in order, set up a proportion, and solve. Check that all lengths use the same units.

**23.1** FRQ · show your working

Identify the similar triangles. Then find x.

Reveal · p. 139 · 5

**(a)** State corresponding vertices and show a proportion.

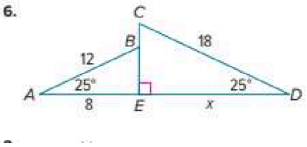



**23.2** FRQ · show your working

Identify the similar triangles. Then find x.

Reveal · p. 139 · 6

**(a)** State corresponding vertices and show a proportion.



02 / WRITE THE STEPS

# Q23 · Written practice 2

## Q23 Solving with similar triangles

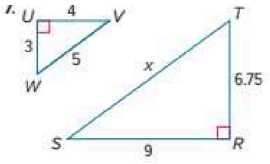
حل المسائل باستعمال المثلثات المتشابهة

**Remember** Establish similarity by AA, SAS, or SSS. Write corresponding vertices in order, set up a proportion, and solve. Check that all lengths use the same units.

**23.3** FRQ · show your working

Identify the similar triangles. Then find x.

Reveal · p. 139 · 7

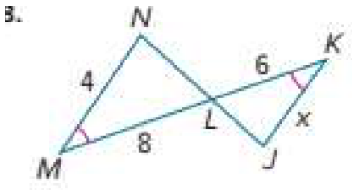



**(a)** State corresponding vertices and show a proportion.

**23.4** FRQ · show your working

Identify the similar triangles. Then find x.

Reveal · p. 139 · 8




**(a)** State corresponding vertices and show a proportion.

02 / WRITE THE STEPS

# Q24 · Written practice 1

Q24

## Distance and midpoint in space

المسافة ومنتصف القطعة في الفراغ

**Remember** Distance = $\sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]}$. Midpoint = $((x_1 + x_2)/2, (y_1 + y_2)/2, (z_1 + z_2)/2)$.

**24.1** FRQ · show your working

Find the midpoint of the segment joining the two points.

$$K(-2, -4, -4), \quad L(4, 2, 0)$$

Reveal · p. 177 · 13

**(a)** Determine the coordinates of midpoint M.

**24.2** FRQ · show your working

Find the midpoint of the segment joining the two points.

$$R(3, 3, 4), \quad V(5, 4, 13)$$

Reveal · p. 177 · 15

**(a)** Determine the coordinates of midpoint M.

02 / WRITE THE STEPS

# Q24 · Written practice 2

Q24

## Distance and midpoint in space

المسافة ومنتصف القطعة في الفراغ

**Remember** Distance = $\sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]}$. Midpoint = $((x_1 + x_2)/2, (y_1 + y_2)/2, (z_1 + z_2)/2)$.

**24.3** FRQ · show your working

Find the distance and midpoint of the segment.

$$P(-5, -2, -1), \quad Q(-1, 0, 3)$$

Reveal · p. 178 · 19

**(a)** Determine the distance.

**(b)** Determine the coordinates of midpoint M.

**24.4** FRQ · show your working

Find the distance and midpoint of the segment.

$$B(\sqrt{3}, 2, 2\sqrt{2}), \quad C(-2\sqrt{3}, 4, 4\sqrt{2})$$

Reveal · p. 178 · 23

**(a)** Determine the distance.

**(b)** Determine the coordinates of midpoint M.

02 / WRITE THE STEPS

# Q25 · Written practice 1

Q25

## Law of Sines and the ambiguous case

قانون الجيوب والحالة المبهمة

**Remember** Use a/sin A = b/sin B. For SSA, sin B = b sin A/a. Test both $B_1 = \sin^{-1}(\text{value})$ and $B_2 = 180° - B_1$, accepting only cases with A + B < 180°. Opposite sides and angles share letters.

**25.1** FRQ · show your working

Determine whether the triangle has no solution, one solution or two solutions. Then solve it. Round sides to the nearest tenth and angles to the nearest degree.

$$A = 50°, \quad a = 34, \quad b = 40$$

Reveal · p. 208 · 15

**(a)** Determine the number of solutions and justify it.

**(b)** Give every possible triangle, or explain why none exists.

**25.2** FRQ · show your working

Determine whether the triangle has no solution, one solution or two solutions. Then solve it. Round sides to the nearest tenth and angles to the nearest degree.

$$A = 24°, \quad a = 3, \quad b = 8$$

Reveal · p. 208 · 16

**(a)** Determine the number of solutions and justify it.

**(b)** Give every possible triangle, or explain why none exists.

02 / WRITE THE STEPS

# Q25 · Written practice 2

Q25

## Law of Sines and the ambiguous case

قانون الجيوب والحالة المبهمة

**Remember** Use a/sin A = b/sin B. For SSA, sin B = b sin A/a. Test both $B_1 = \sin^{-1}(\text{value})$ and $B_2 = 180° - B_1$, accepting only cases with A + B < 180°. Opposite sides and angles share letters.

**25.3** FRQ · show your working

Determine whether the triangle has no solution, one solution or two solutions. Then solve it. Round sides to the nearest tenth and angles to the nearest degree.

$$A = 125°, \quad a = 22, \quad b = 15$$

Reveal · p. 208 · 17

**(a)** Determine the number of solutions and justify it.

**(b)** Give every possible triangle, or explain why none exists.

**25.4** FRQ · show your working

Determine whether the triangle has no solution, one solution or two solutions. Then solve it. Round sides to the nearest tenth and angles to the nearest degree.

$$A = 30°, \quad a = 2, \quad b = 4$$

Reveal · p. 208 · 19

**(a)** Determine the number of solutions and justify it.

**(b)** Give every possible triangle, or explain why none exists.

03 / CHECK & LEARN

# MCQ answer key

Check after solving. Match the printed question number to its correct option. Full explanations are available in the online LO bank.

**Q1 · Polynomial operations**
1.1: C 1.2: B 1.3: A 1.4: D

**Q2 · FOIL and binomial products**
2.1: B 2.2: A 2.3: D 2.4: C

**Q3 · Factoring by GCF and grouping**
3.1: A 3.2: D 3.3: C 3.4: B

**Q4 · Factoring monic trinomials**
4.1: D 4.2: C 4.3: B 4.4: A

**Q5 · Interior angles of polygons**
5.1: C 5.2: B 5.3: A 5.4: D

**Q6 · Properties of parallelograms**
6.1: B 6.2: A 6.3: D 6.4: C

**Q7 · Tests for parallelograms**
7.1: A 7.2: D 7.5: A 7.6: C

**Q8 · Kites**
8.1: D 8.2: C 8.3: B 8.4: A

**Q9 · Dilation coordinates**
9.1: C 9.2: B 9.3: A 9.4: D

**Q10 · Similar polygons**
10.1: B 10.2: A 10.3: D 10.4: C

**Q11 · Dilation scale factors**
11.1: A 11.2: D 11.3: C 11.4: B

**Q12 · AA triangle similarity**
12.1: D 12.2: C 12.5: D 12.6: A

**Q13 · Proportional parts in triangles**
13.1: C 13.2: B 13.3: A 13.4: D

**Q14 · Parts of similar triangles**
14.1: B 14.2: A 14.3: D 14.4: C

**Q15 · Geometric mean**
15.1: A 15.2: D 15.3: C 15.4: B

**Q16 · Pythagorean theorem and converse**
16.1: D 16.2: C 16.5: C 16.6: C

**Q17 · Special right triangles**
17.1: C 17.2: B 17.5: B 17.6: A

**Q18 · Right-triangle trigonometry**
18.1: B 18.2: A 18.3: D 18.4: C

**Q19 · Angles of elevation and depression**
19.1: A 19.2: D 19.3: C 19.4: B

**Q20 · Area with two sides and an angle**
20.1: D 20.2: C 20.3: B 20.4: A

03 / CHECK & LEARN

# Worked solutions 1

**21.1** p. 680 · 35

**(a)** Extract any greatest common factor, then use the difference-of-squares or perfect-square identity. Correct method (5), complete factors and check (5).

$$8\left(-3p+x\right)\left(3p+x\right)$$

**21.2** p. 680 · 38

**(a)** Extract any greatest common factor, then use the difference-of-squares or perfect-square identity. Correct method (5), complete factors and check (5).

$$4\left(2b-5\right)\left(2b+5\right)$$

**21.3** p. 681 · 51

**(a)** Yes; Extract any greatest common factor, then use the difference-of-squares or perfect-square identity. Correct method (5), complete factors and check (5).

$$\left(3z-1\right)^{2}$$

**21.4** p. 681 · 52

**(a)** Yes; Extract any greatest common factor, then use the difference-of-squares or perfect-square identity. Correct method (5), complete factors and check (5).

$$\left(6x-5\right)^{2}$$

**22.1** p. 64 · 9

**(a)** Exterior angle=180−144=36°. Therefore n=360/36=10. Correct exterior angle (3), equation (3), number of sides (4).

$$n=10$$

**22.2** p. 64 · 11

**(a)** Exterior angle=180−160=20°. Therefore n=360/20=18. Correct exterior angle (3), equation (3), number of sides (4).

$$n=18$$

**22.3** p. 64 · 15

**(a)** The exterior angles sum to 360° (3). Combine all labelled exterior angles: 9x+27=360 (4), so x=37 (3).

$$x=37$$

**22.4** p. 64 · 18

**(a)** The exterior angles sum to 360° (3). Combine all labelled exterior angles: 4x+76=360 (4), so x=71 (3).

$$x=71$$

**23.1** p. 139 · 5

**(a)** The vertical angles at J are equal and H,K are right angles. 8/4=10/x gives x=5. Correct ordered correspondence (4), proportion (3), x (3).

$$\triangle IHJ\sim\triangle LKJ,\quad x=5$$

**23.2** p. 139 · 6

**(a)** Both have a right angle at E and a marked 25° angle. 12/18=8/x gives x=12. Correct ordered correspondence (4), proportion (3), x (3).

$$\triangle ABE\sim\triangle DCE,\quad x=12$$

03 / CHECK & LEARN

# Worked solutions 2

### 23.3 p. 139 · 7

**(a)** The ratios 9/4=6.75/3=2.25 agree. The hypotenuse x=5·2.25=11.25. Correct ordered correspondence (4), proportion (3), x (3).

$$\triangle UVW \sim \triangle RTS, \quad x = \frac{45}{4}$$

### 23.4 p. 139 · 8

**(a)** Use the marked angles at M,K and vertical angles at L. 4/x=8/6 gives x=3. Correct ordered correspondence (4), proportion (3), x (3).

$$\triangle MNL \sim \triangle KJL, \quad x = 3$$

### 24.1 p. 177 · 13

**(a)** Average the corresponding x, y and z coordinates separately. Correct component calculations and complete midpoint.

$$M = (1, -1, -2)$$

### 24.2 p. 177 · 15

**(a)** Average the corresponding x, y and z coordinates separately. Correct component calculations and complete midpoint.

$$M = (4, \tfrac{7}{2}, \tfrac{17}{2})$$

### 24.3 p. 178 · 19

**(a)** Use $d=\sqrt{[(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2]}$; substitution (3) and simplified exact distance (2).

$$d = 6$$

**(b)** Average the corresponding x, y and z coordinates separately. Correct component calculations and complete midpoint.

$$M = (-3, -1, 1)$$

### 24.4 p. 178 · 23

**(a)** Use $d=\sqrt{[(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2]}$; substitution (3) and simplified exact distance (2).

$$d = \sqrt{39}$$

**(b)** Average the corresponding x, y and z coordinates separately. Correct component calculations and complete midpoint.

$$M = (-\tfrac{\sqrt{3}}{2}, 3, 3\sqrt{2})$$

### 25.1 p. 208 · 15

**(a)** sin B=b sin A/a=0.90122876. Two solutions; test both arcsine candidates against A+B<180°. Correct ratio (2), count and admissibility (2).

**(b)** B≈64°, C≈66°, c≈40.4; B≈116°, C≈14°, c≈11.0 Calculate with full-precision angles before rounding. Correct possible triangles and remaining side(s) (6).

### 25.2 p. 208 · 16

**(a)** sin B=b sin A/a=1.08463105. No solution; test both arcsine candidates against A+B<180°. Correct ratio (2), count and admissibility (2).

**(b)** sin B>1, so there is no real angle B and no triangle. Calculate with full-precision angles before rounding. Correct possible triangles and remaining side(s) (6).

### 25.3 p. 208 · 17

**(a)** sin B=b sin A/a=0.55851276. One solution; test both arcsine candidates against A+B<180°. Correct ratio (2), count and admissibility (2).

**(b)** B≈34°, C≈21°, c≈9.6 Calculate with full-precision angles before rounding. Correct possible triangles and remaining side(s) (6).

### 25.4 p. 208 · 19

**(a)** sin B=b sin A/a=1.00000000. Two solutions; test both arcsine candidates against A+B<180°. Correct ratio (2), count and admissibility (2).

**(b)** B≈90°, C≈60°, c≈3.5; B≈90°, C≈60°, c≈3.5 Calculate with full-precision angles before rounding. Correct possible triangles and remaining side(s) (6).